

# Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Yellow-breasted Chat"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

**Check whether MCMC worked:**

**Do we have divergent transitions?**

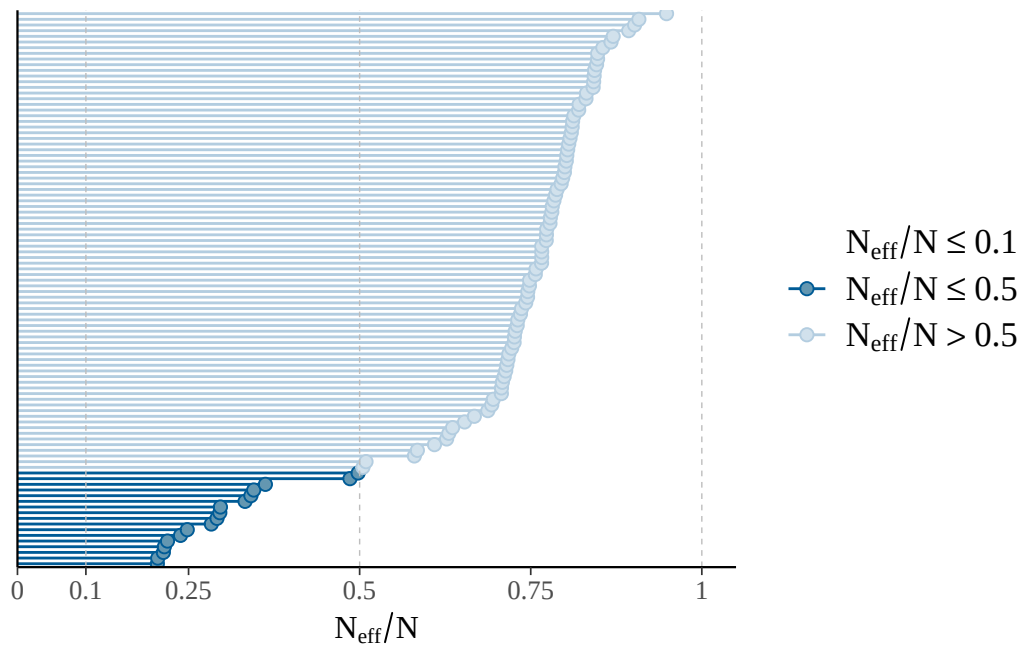
```
[1] "divergent transitions: 0"
```

**Explorations regarding divergent transitions:**

...

**Is effective number of MCMC samples large enough:**

The larger the ratio of  $n_{\text{eff}}$  to  $N$  (number of samples), the better.



```
[1] "effective sample size fine"
```

```
[1] "n_eff ratio bulk ESS:"
```

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.2044 | 0.7304  | 0.9595 | 0.9106 | 1.1020  | 2.0186 |

```
[1] "n_eff ratio tail ESS:"
```

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.3303 | 0.6933  | 0.7561 | 0.7145 | 0.8030  | 0.9484 |

```
[1] "effective sample size in bulk and tail fine"
```

### **Do chains mix well:**

No divergences to plot.

No divergences to plot.



No divergences to plot.





Post-warmup iteration

### Are R-hat values fine:

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.9992 | 0.9997  | 1.0001 | 1.0003 | 1.0004  | 1.0046 |

```
[1] "All R-hat values < 1.01"
```

### Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_Intercept"      "b_occ_bio1_3yrs"
[3] "b_occ_pr_autumn_3yrs" "b_occ_pr_winter_3yrs"
[5] "b_occ_lbio1_3yrsE2"   "b_occ_lpr_spring_3yrsE2"
[7] "b_occ_urban_3yrs"
```

```
[1] "b_colo_bio1"          "b_colo_pr_autumn"      "b_colo_lbio1E2"
[4] "b_colo_lbio3E2"       "b_colo_lpr_springE2"   "b_colo_lpr_winterE2"
[7] "b_colo_pastr"         "b_colo_urban"          "b_colo_lsecdfe2"
[10] "b_colo_lpastrE2"      "b_colo_lurbanE2"
```

```
[1] "b_ex_bio1"           "b_ex_pr_spring"
[3] "b_ex_pr_winter"      "b_ex_lbio2E2"
[5] "b_ex_lbio7E2"        "b_ex_lpr_springE2"
[7] "b_ex_lpr_summerE2"   "b_ex_lpr_winterE2"
[9] "b_ex_urban"          "b_ex_lsum_annual_cropsE2"
```

